

PRODUCT REQUIREMENTS DOCUMENT

NovalD: National Identity Civil Registration System (NICRS)

PRD Title	NovalD: National Identity Civil Registration System (NICRS)
Managed By	Novaria Identity and Civil Registration Authority (NICRA), an agency under the Novaria Ministry of Interior
Version	1.0
Product Manager	Gio
Product Status	Conceptual product for demonstration and learning purposes
Platform	Web-based national identity and civil registration platform, with a citizen-facing “Log in with NovalD” authentication service
Technical Architecture	Proposed cloud-based, modular architecture
Companion Reference	Novaria Administrative Reference.xlsx: canonical states, zones and LGAs shared across all Novaria products

Security Notice: This document describes a fictional system created for demonstration and learning purposes. Novaria, NICRA, NovalD, the Novaria Social Number and all sample information are fictional.

1. BACKGROUND AND PURPOSE

Novaria currently lacks a single, authoritative system for registering citizens and managing their identities throughout their lifetime. Birth records, naturalization records, identity information and death records are maintained across different institutions. This makes it difficult to establish and maintain one reliable identity for each citizen.

NovalD is a national identity and civil registration platform managed by the Novaria Identity and Civil Registration Authority (NICRA). It will create and maintain a trusted identity record for every eligible citizen and assign each citizen a unique Novaria Social Number (NSN).

The NSN will serve as the common identity reference across approved government services. Systems such as OneHealth will verify a person's identity through NovalD without receiving unrelated personal information.

NovalD will also serve as the public-facing identity platform through which citizens can access approved digital government services. Where enabled, citizens will be able to select **“Log in with NovalD”** on a connected government platform and authenticate without creating a separate identity for each service. NICRA remains the government authority responsible for regulating, operating and overseeing the NovalD platform.

This document uses three related names precisely, and keeps them distinct on purpose. **NICRA** (Novaria Identity and Civil Registration Authority) is the government agency responsible for the system: use it wherever the document is talking about who runs, approves or governs something. **NICRS** (National Identity Civil Registration System) is the formal, technical name of the system itself; it appears in this document's title and belongs in technical, legal or architectural language. **NovalD** is the public-facing brand a citizen actually sees and interacts with, such as on a login screen; use it for citizen-facing flows and everyday references throughout the rest of this document.

The NSN is different again, and not a fourth name for the same thing: it is the 12-digit identifier **NovalD** issues to each citizen, the way a passport office issues passport numbers rather than being one itself. A citizen **has** an NSN; they **log in with** NovalD, not with an NSN.

NovalD will support the full identity lifecycle, including:

- Birth notification and registration.
- Registration of naturalized citizens.
- Issuance of an NSN.
- Identity verification.
- Identity updates and corrections.
- Periodic photo and biometric updates where applicable.
- Death notification and registration.
- Secure verification by approved government institutions.

1.1 Objectives

- Establish one authoritative identity record for each citizen.
- Register births and naturalized citizens through standardized processes.
- Generate and manage a unique NSN for every eligible citizen.
- Reduce duplicate, fraudulent and conflicting identity records.
- Enable secure identity verification across approved government systems.
- Maintain a complete audit history of identity-related actions.

- Improve the reliability of national population information.
- Protect citizen information through role-based and purpose-based access.
- Provide a trusted identity foundation for OneHealth and future Novarian government systems.
- Provide a secure “Log in with NovalD” service for approved government digital platforms.
- Reduce the need for citizens to create and maintain separate identities across multiple government services.

2. PROBLEM STATEMENT

Identity information in Novaria is fragmented, inconsistent and difficult to verify across institutions. Government agencies maintain separate records that are not always linked or synchronized. This fragmentation creates several challenges:

- Citizens may have multiple or conflicting identities.
- Birth and death records may be incomplete.
- Identity issuance may be delayed because information cannot be verified quickly.
- Government institutions may struggle to confirm that an individual is who they claim to be.
- Duplicate and fraudulent registrations may go undetected.
- Citizens may be required to submit the same information repeatedly to different institutions.
- Government planning may rely on incomplete population information.
- Personal information may be shared excessively or without adequate controls.
- Citizens may have difficulty accessing essential services when their identities cannot be verified.
- Changes made in one government system may not be reflected in other connected systems.

NovalD will address these issues by establishing a centralized national identity registry, standardized civil registration processes and controlled identity-verification services.

3. PRODUCT PRINCIPLES

3.1 One Person, One Identity

Each citizen will have one active NovalD identity record and one NSN. The NSN will not be reassigned to another person.

3.2 Privacy by Design

Connected institutions will only receive the minimum information required for an approved purpose.

3.3 Authoritative Identity Source

NovalD will be the official source of core citizen identity information. Connected systems may maintain their own service-specific records but should use the NSN to identify the citizen.

3.4 Traceability

Every identity registration, search, verification, update, approval, rejection and exceptional access must be recorded.

3.5 Human Oversight

High-risk registrations, suspected duplicates, material identity changes and disputed records must be reviewed by authorised officers.

3.6 Inclusive Access

NovalD will support online and assisted registration. Controlled low-connectivity processes will be available for approved centres in areas with unreliable internet access.

3.7 Separation of Government Records

NovalD will not store complete medical, criminal, educational or social-service case records. These will remain within the systems responsible for them.

3.8 No Silent Identity Changes

Material changes to a citizen's identity must be supported by evidence, independently approved and communicated to the citizen where appropriate.

4. GOALS AND SUCCESS METRICS

#	Metric	Baseline	Year-One Target
1	Approved registration centres onboarded	0	1,000+
2	Eligible registrations assigned an NSN	0	95%
3	New births registered within the statutory period	To be established	85%
4	Standard identity-verification response time	Not available	Five seconds or less
5	Potential duplicate registrations routed for review	Not available	100%
6	High-risk identity changes independently approved	Not available	100%
7	Critical activities captured in audit logs	Not available	100%
8	Core platform availability	Not available	99.9%
9	Standard registration applications completed within the agreed service period	Not available	90%
10	Connected institutions completing periodic access reviews	0	100%

Novaria's population is set at approximately 38,000,000 for planning purposes, a figure established for this PRD rather than measured (recorded in the Platform Reference Decisions Log). Year-One targets above assume an initial enrolment wave of roughly 5% of the population, approximately 1,900,000 registrations, not universal coverage in year one; NICRA will revise both the population figure and these targets once real registration volumes are available.

5. GOVERNANCE AND INSTITUTIONAL OWNERSHIP

5.1 Novaria Identity and Civil Registration Authority

NICRA will manage NovalD under the oversight of the Novaria Ministry of Interior. NICRA will be responsible for:

- Implementing national identity and civil registration policies.
- Approving registration centres.
- Authorising institutions to access NovalD.
- Issuing and managing NSNs.
- Maintaining the core national identity registry.
- Reviewing identity corrections and disputes.
- Managing potential duplicate records.
- Investigating suspected registration fraud.
- Monitoring access to citizen information.
- Establishing identity-verification standards.

- Managing audit, compliance and operational reporting.
- Suspending users, centres or institutions that misuse the system.

5.2 Connected Institutions

OneHealth

OneHealth will submit birth notifications for births occurring in connected hospitals and health facilities. OneHealth will:

- Capture birth information at the point of care.
- Submit the notification to NovalD.
- Receive a unique birth-notification reference.
- Track the notification status.
- Correct rejected or incomplete notifications where permitted.
- Receive confirmation when the birth has been formally registered.

OneHealth will not approve civil registration or generate an NSN. Those responsibilities remain with NICRA. Births occurring outside OneHealth-connected facilities must be supported through civil registration centres and other approved notification channels.

Civil Registration Centres

- Register births.
- Process births that occurred outside health facilities.
- Process late birth registrations.
- Capture identity information and supporting evidence.
- Capture photographs and biometrics where applicable.
- Provide assisted services.
- Initiate approved identity updates.
- Submit applications to NICRA for review and approval.

Novaria Citizenship and Immigration Service

The Immigration authority will:

- Assess citizenship and naturalization applications.
- Approve or reject citizenship.
- Provide validated naturalization references to NovalD.
- Confirm the authenticity and status of naturalization approvals.

NovalD will not decide whether a person qualifies for citizenship. NICRA will create the identity and issue an NSN after Immigration has approved citizenship and NovalD checks have been completed.

Police Services

The Police system may use NovalD for lawful identity verification. Police access will be limited by role, purpose and legal authority. Criminal investigations, arrests, warrants, convictions, missing-person cases and police intelligence will remain within a separate Police Case and Investigation Management System.

Social Services

Social Services may use the NSN to identify children and other persons receiving government support. Child-protection assessments, foster-care information and social-service case notes will remain in the Social Services system.

Other Government Institutions

Additional government institutions may be connected after completing:

- Business-purpose approval.
- Data-protection review.
- Security assessment.
- Integration testing.
- Access configuration.
- User training.
- Formal onboarding by NICRA.

6. USER ROLES AND PERMISSIONS

Role	Access Scope	Core Permissions
NICRA National Administrator	Nationwide	Manage institutions, centres, roles, access policies and national system configuration
NICRA Regional Administrator	Assigned region	Monitor centres, officers and applications within the assigned region
Registration Centre Administrator	Assigned centre	Manage centre staff and monitor centre operations
Registration Officer	Assigned centre	Create and submit registrations and correction requests
Verification Officer	Assigned queue or region	Review information, validate evidence and request clarification
Senior Approver	Assigned region or nationwide	Approve high-risk registrations, material corrections and duplicate resolutions
Duplicate Review Officer	Assigned review queue	Investigate possible duplicate identities
Birth Notification Officer	Approved health facility	Submit and track birth notifications
Immigration Officer	Approved Immigration unit	Submit and confirm naturalization approval references
Government Verification User	Approved institution	Verify permitted identity attributes for an approved purpose
Compliance Officer	Assigned oversight scope	Review access history, unusual activity and compliance reports
Auditor	Read-only audit scope	Review audit records without modifying identity information
Citizen	Personal access	View permitted personal information and initiate requests
System Support Administrator	Technical operations	Monitor system health without unrestricted access to readable identity records

6.1 Permission Rules

- A user must only access information required for their assigned role.
- Users must not share accounts.
- Registration Officers must not approve applications they created.
- Verification Officers must not approve high-risk changes where independent approval is required.
- Government Verification Users must select or provide an approved verification purpose.
- Sensitive fields must be masked when they are not required.
- System Support Administrators must not have routine access to readable citizen records.
- Emergency access must be time-limited, justified and reviewed.

- Suspended users must lose access immediately.
- Historical actions must remain in the audit log after a user is suspended or removed.
- Access must be reviewed periodically by NICRA and the connected institution.

7. FUNCTIONAL REQUIREMENTS

7.1 Birth Notification from OneHealth

For a birth occurring in a connected health facility, OneHealth must submit:

- Child's first name where already provided.
- Child's surname or proposed surname.
- Date and time of birth.
- Place of birth.
- Sex recorded at birth.
- Birth type, such as single, twin or triplet.
- Birth order for multiple births.
- Mother's name and NSN where available.
- Father's name and NSN where available.
- Guardian information where applicable.
- Attending healthcare professional.
- Health facility.
- Date and time of notification.
- Supporting birth record reference.

NovalD must:

- Validate that required fields are present.
- Check that the health facility and submitting officer are authorised.
- Generate a unique birth-notification reference.
- Check for a possible earlier notification of the same birth.
- Return the notification status to OneHealth.
- Place incomplete or conflicting notifications in an exception queue.
- Prevent the birth notification from automatically generating an NSN.

7.2 Birth Registration

A birth notification becomes a formal identity record only after civil registration is completed. The Registration Officer must:

1. Retrieve the birth notification or create a registration for a birth without an electronic notification.
2. Confirm the parent or guardian information.
3. Capture required residential and contact information.
4. Review supporting evidence.
5. Capture a child photograph where required by policy.
6. Submit the application for validation.

NovalD must:

- Check for possible duplicate births or identities.
- Validate parent or guardian NSNs where provided.
- Flag conflicting names, dates or parent information.

- Route standard applications to a Verification Officer.
- Route high-risk or late registrations for additional review.
- Generate the NSN only after final approval.
- Preserve the registration source and supporting evidence.
- Notify the parent or guardian of the outcome.

7.3 Birth Outside a Health Facility

Where a birth occurs outside a registered health facility, NovalD must allow an authorised Registration Officer to initiate the registration. The application must include:

- Details of the child.
- Parent or guardian information.
- Place and circumstances of birth.
- Supporting evidence required by NICRA: typically a signed statement from the attending health worker or traditional birth attendant, or, where none was present, a sworn affidavit.
- Witness or community verification where permitted.
- Reason no hospital notification is available.

Such applications must be assigned a higher review level than a standard hospital-notified birth.

7.4 Late Birth Registration

NovalD must classify a registration as late when it is submitted more than 60 days after the date of birth. Late registration must:

- Capture the reason for the delay.
- Require additional supporting evidence, such as a school enrolment record, a religious or baptismal record, or a sworn affidavit of age declaration.
- Perform enhanced duplicate checks.
- Require approval by an authorised senior officer.
- Preserve all evidence and decisions.
- Allow the applicant to appeal a rejection.

7.5 Naturalized Citizen Registration

Naturalized citizen registration must begin only after Immigration has approved citizenship. NovalD must:

- Validate the naturalization approval reference.
- Confirm that the approval is active and has not been withdrawn.
- Capture the person's legal name.
- Capture previous names.
- Capture date and place of birth.
- Capture previous nationality where required.
- Capture contact and residential information.
- Capture photograph and approved biometrics.
- Check for an existing NSN or identity record.
- Route possible matches to a Duplicate Review Officer.

- Generate an NSN after successful approval.
- Retain the Immigration reference without exposing restricted immigration records.

7.6 Citizen Identity Record

The core identity record will contain:

Primary Identity Information

- NSN.
- Legal first name.
- Middle name.
- Surname.
- Previous legal names.
- Date of birth.
- Place of birth.
- Sex recorded at registration.
- Citizenship status.
- Basis of citizenship.
- Registration date and source.
- Identity status.

Relationship Information

- Mother's name and NSN where available.
- Father's name and NSN where available.
- Legal guardian information where applicable.
- Relationship status and supporting authority.

Contact Information

- Residential address.
- State and district.
- Phone number.
- Email address where available.
- Preferred notification channel.

Identity Attributes

- Current photograph.
- Previous photographs.
- Approved biometric references.
- Physical identification credential status where applicable.

Record Information

- Registration centre.
- Creating officer.
- Approval date.
- Record version.
- Previous values.

- Correction and update history.
- Date and place of death where applicable.

7.7 Mandatory and Optional Information

Mandatory information must depend on the registration type and age of the applicant. NovalD must:

- Clearly identify mandatory fields.
- Prevent submission when required information is missing.
- Permit approved exceptions where information is genuinely unavailable.
- Record the reason for every exception.
- Distinguish between information that is unknown and information that is not applicable.

The detailed mandatory and optional fields will be documented in the Data Dictionary appendix.

7.8 Photo and Biometric Lifecycle

Biometric information forms part of the NovalD MVP because demographic information alone may not be sufficient to prevent duplicate identities in a large population.

Biometrics Included in the MVP

- Facial photograph.
- Facial biometric template.
- Fingerprints for eligible citizens.
- Biometric exception records for citizens whose biometrics cannot be captured reliably.

Iris scanning will not form part of the MVP. It may be considered in a future version if required.

Biometric Requirements for Children

- A photograph may be captured at initial registration where required.
- The identity will initially be linked to the verified parent or guardian records and the registered birth event.
- The child's photograph must be updated at age five.
- Fingerprints will be enrolled from age five where they can be captured reliably.
- Biometrics must be updated again at ages 12 and 18 as the child develops.
- NICRA must provide advance notification and a reasonable grace period for mandatory updates.

The absence of usable fingerprints must not prevent an eligible child from receiving an NSN. Failure to update a photograph must not automatically cancel citizenship or delete the NSN.

Biometric Requirements for Adults

Adults registering through naturalization, late registration or another approved enrolment process must provide a current facial photograph, a facial biometric template, and available fingerprints captured according to NICRA standards.

Adults must recapture their facial photograph, facial biometric template and available fingerprints at intervals set by NICRA, nominally every ten years, or sooner if requested or if existing biometrics can no longer be matched reliably.

Where fingerprints cannot be captured because of disability, injury, age or another valid reason:

- The Registration Officer must record the exception.
- Supporting information may be required.
- The application must be routed for enhanced review.
- Alternative identity-verification methods must be used.

Biometric Protection

Biometric information must:

- Be encrypted in transit and at rest.
- Be stored separately from ordinary demographic data where technically possible.
- Be converted into protected biometric templates for matching.
- Be accessible only to authorised users and services.
- Not be displayed in ordinary search results.
- Not be included in standard reports or exports.
- Be retained and used only according to approved NICRA policies.
- Have every access and matching request recorded in the audit log.

7.9 NSN Generation and Management

The Novaria Social Number will use the following format:

- The NSN will contain 12 numeric digits.
- The first 11 digits will be generated randomly.
- The final digit will be a check digit used to detect common entry errors.
- The number will be displayed in groups of four for readability: XXXX-XXXX-XXXX.
- Dashes used for display will not form part of the stored NSN.
- The number must not contain embedded information about the citizen.

The NSN must not reveal:

- Date of birth.
- Sex.
- State or district.
- Registration centre.
- Citizenship type.
- Registration date.

The NSN-generation service must:

- Use a secure random-number generation process.
- Check the complete number against all existing NSNs before issuance.
- Apply the approved check-digit algorithm (Luhn / mod-10).
- Reject and regenerate any duplicate number.
- Record the date, time and system event responsible for generation.
- Prevent users from generating an NSN manually.
- Prevent an issued NSN from being changed or reused.
- Generate an NSN only after the registration has received final approval.

Where a citizen forgets their NSN, NovalD must retrieve the existing number after successful identity verification; a replacement NSN must not be created. Where a physical or digital

credential is lost or compromised, the credential may be revoked and replaced, but the citizen's NSN must remain unchanged, and the incident must be logged.

7.10 Log in with NovalD: Digital Authentication Service

NovalD provides an authentication service known as **“Log in with NovalD.”** Approved government platforms may use this service to authenticate citizens without creating separate identity credentials for each platform.

The authentication process must:

1. Redirect the citizen from the connected service to NovalD.
2. Display the name of the requesting service.
3. Display the information the service is requesting.
4. Authenticate the citizen using an approved method.
5. Confirm that the requesting service is authorised.
6. Return an authentication result and only the approved identity attributes.
7. Record the authentication event in the audit log.
8. Return the citizen to the requesting service.

Connected services must not receive the citizen's NovalD password, PIN or biometric information. Authentication methods may include:

- Password and one-time password.
- NovalD PIN.
- Approved authenticator application.
- Device-based authentication.
- Biometric authentication performed on the citizen's device.

The exact authentication methods included in the MVP will be defined during security design.

7.11 Identity Search

NovalD must support authorised searches using approved combinations such as:

- NSN.
- Full name and date of birth.
- Name and registered phone number.
- Name and parent or guardian details.
- Approved biometric match for authorised officers.

Search controls must include:

- Minimum search criteria.
- Role-based result limits.
- Masking of sensitive information.
- Rate limits.
- Prevention of unrestricted browsing.
- Logging of the search purpose.
- Monitoring of repeated searches for unrelated citizens.
- Additional approval for bulk or exceptional searches.

7.12 Identity Verification

Connected institutions may submit selected identity attributes for verification. NovalD may return:

- Match.
- Partial match.
- No match.
- Record under review.
- Record restricted.
- Deceased status where the requesting institution is authorised to receive it.
- Service temporarily unavailable.

NovalD must not return the complete identity record when a simple verification result is sufficient.

7.13 Identity Updates and Corrections

NovalD must distinguish between:

Minor Updates

Examples include: phone number, email address, residential address. Minor updates may be completed after standard authentication and validation.

Material Identity Changes

Examples include: legal name, date of birth, place of birth, parentage, citizenship information, sex recorded in the identity record, deceased status, and linking or merging identity records.

Material changes must:

- Require evidence appropriate to the specific change, for example a court order, marriage certificate or government gazette notice for a legal name change, or medical or school records plus a sworn affidavit for a date-of-birth correction.
- Be reviewed by a Verification Officer.
- Be approved by a separate Senior Approver.
- Preserve the previous value.
- Record the reason for the change.
- Record the initiating and approving officers.
- Notify the citizen where appropriate.
- Send approved updates to affected connected systems.

7.14 Duplicate Detection and Resolution

NovalD must screen every new registration against existing identity records before an NSN is issued. Duplicate checks will use:

- Legal and previous names.
- Date and place of birth.
- Parent or guardian information.
- Contact information.
- Supporting-document references.
- Facial biometric matching.

- Fingerprint matching where fingerprints are available.

The system must not automatically merge or reject records based only on a biometric score. Potential matches must:

1. Be placed in a dedicated review queue.
2. Temporarily pause NSN issuance.
3. Display the demographic and biometric factors contributing to the match.
4. Be reviewed by an authorised Duplicate Review Officer.
5. Allow additional evidence to be requested.
6. Require an independently approved decision.
7. Preserve the original records and decision history.
8. Allow the affected citizen to dispute an incorrect match.

NICRA will define separate matching thresholds for low-confidence matches that may proceed without escalation, medium-confidence matches requiring officer review, and high-confidence matches requiring enhanced investigation. The specific numeric thresholds will be validated during biometric testing because performance may vary by biometric provider, age group and data quality.

7.15 Death Notification and Registration

Death notifications may be received from OneHealth-connected health facilities, other approved health facilities, Civil Registration Centres, approved government authorities, or an authorised family or representative process. NovalD must:

- Validate the deceased person's identity.
- Capture date and place of death.
- Capture the notifier and source.
- Capture supporting evidence.
- Check for conflicting information.
- Require approval before changing the identity status.
- Preserve the identity record and NSN.
- Notify approved connected systems.
- Prevent prohibited use of the deceased person's identity.
- Support urgent reversal where a person has been incorrectly marked as deceased.

An incorrect death registration must be treated as a critical incident.

7.16 Citizen Self-Service

Where included in the approved release, citizens may:

- Sign in securely.
- View permitted identity information.
- View their NSN.
- View and self-print a virtual identity card.
- Track registration and correction requests.
- Submit minor update requests.
- Submit correction requests with evidence.
- Report suspected identity misuse.

- Receive notifications.
- View selected access history where permitted.

Citizens must not be able to directly edit material identity fields.

7.17 Registration Centre Management

NICRA must be able to:

- Create and approve a registration centre.
- Assign the centre to a state and LGA using the canonical administrative structure (see Section 20).
- Define permitted services.
- Assign a Centre Administrator.
- Approve centre staff.
- Monitor registrations and exceptions.
- Suspend or deactivate the centre.
- Preserve historical centre activities.
- Transfer unresolved applications to another centre where necessary.

7.18 Institutional Access Management

NICRA must be able to:

- Register an institution.
- Record its approved business purposes.
- Define accessible identity attributes.
- Set transaction and search limits.
- Issue integration credentials.
- Set credential expiry dates.
- Suspend or revoke credentials.
- Monitor institution activity.
- Require periodic access certification.
- Record the institution's data-protection and security approvals.

7.19 Notifications

NovalD must send notifications for: birth notification submission, registration submission, requests for additional evidence, approval, rejection, NSN issuance, correction-request status, material identity changes, photo or biometric update reminders, suspected identity misuse, and institutional or user suspension.

Supported channels may include in-platform notification, email, SMS, and approved government notification channels. Sensitive personal information must not be included in unsecured notifications.

Failed notifications must be retried according to configured rules, be recorded, be available in an operational failure queue, and must not reverse an otherwise successful identity action.

7.20 Identifiers for Non-Citizens

Lawful residents and refugees who are not Novaria citizens are not issued an NSN. NICRA issues these individuals a Novaria Resident Number (NRN): a 12-digit identifier in the same XXXX-XXXX-XXXX display format as the NSN, generated by the same secure random-number process and validated with the same check-digit algorithm, but held in a separate number series so an NRN can never be mistaken for or reused as an NSN.

Connected systems must treat the NRN as a distinct identifier type, not a variant of the NSN, and must record which type an individual holds. Where a resident is later granted citizenship through naturalization, NICRA issues a new NSN and retires the NRN, preserving a record of the change; the individual's prior identity record is linked but not deleted.

8. KEY USER JOURNEYS

8.1 Birth in a OneHealth-Connected Facility

1. The hospital records the birth in OneHealth.
2. OneHealth submits the birth notification to NovalD.
3. NovalD validates the hospital, officer and required information.
4. NovalD generates a birth-notification reference.
5. The parent or guardian completes civil registration at an approved centre or through an approved process.
6. A Registration Officer reviews the notification and supporting evidence.
7. NovalD performs duplicate and consistency checks.
8. A Verification Officer reviews the application.
9. Exceptions are resolved or escalated.
10. The application is approved.
11. NovalD creates the identity record and generates an NSN.
12. OneHealth receives confirmation that the birth has been registered.
13. The parent or guardian receives confirmation.

8.2 Birth Outside a Health Facility

1. The parent or guardian visits an approved Registration Centre.
2. The Registration Officer records the birth details.
3. The required supporting evidence and witness information are captured.
4. NovalD checks for an earlier notification or registration.
5. The application is classified for enhanced review.
6. A Verification Officer reviews the evidence.
7. A Senior Approver approves or rejects the application.
8. NovalD creates the identity and issues an NSN after approval.

8.3 Naturalized Citizen Registration

1. Immigration approves the citizenship application.
2. Immigration sends or confirms the approval reference.
3. The citizen completes identity enrolment.
4. NovalD validates the Immigration approval.
5. NovalD performs demographic and biometric duplicate checks.
6. A Verification Officer reviews the application.
7. Any possible duplicate is sent for investigation.
8. The application is approved.
9. NovalD creates the identity and issues an NSN.

8.4 Identity Correction

1. The citizen or authorised officer submits a correction request.

2. NovalD identifies whether it is minor or material.
3. Required evidence is submitted.
4. A Verification Officer reviews the request.
5. Material changes are sent to a Senior Approver.
6. The request is approved, rejected or returned for more information.
7. NovalD preserves the previous value.
8. The active identity record is updated.
9. The citizen and affected connected systems are notified.

8.5 Potential Duplicate

1. NovalD identifies a possible match.
2. The new application is paused.
3. A Duplicate Review Officer compares the records.
4. Additional evidence is requested where necessary.
5. The officer recommends separate identities, a merge, a link or suspected fraud.
6. A Senior Approver reviews the recommendation.
7. NovalD records the decision.
8. The application continues or is rejected based on the outcome.

8.6 Death Registration

1. An approved source submits a death notification.
2. NovalD validates the identity and evidence.
3. Conflicting information is sent for review.
4. An authorised officer approves the death registration.
5. The citizen's status changes to deceased.
6. The NSN and identity remain preserved.
7. Approved connected systems receive the status update.
8. Any subsequent use of the identity may trigger an alert.

8.7 Log in with NovalD (Digital Authentication)

1. A citizen selects “Log in with NovalD” on an approved government platform.
2. The platform redirects the citizen to NovalD.
3. NovalD displays the requesting service and the information it is requesting.
4. The citizen authenticates using an approved method.
5. NovalD confirms the requesting service is authorised.
6. NovalD returns only the approved identity attributes to the service.
7. The authentication event is recorded in the audit log.
8. The citizen is returned to the requesting service, signed in.

9. INTER-AGENCY INTEGRATION

Connected System	Purpose	Permitted Information
OneHealth	Birth notification and patient verification	Birth-notification status, NSN status and approved demographic verification
Immigration	Naturalization confirmation	Approval reference, citizenship status and identity matching
Police System	Lawful identity verification	Minimum permitted identity attributes
Social Services	Identity and guardian verification	NSN status and approved relationship information
Other Government Services	Identity verification	Purpose-specific verification result

Every integration must have an approved purpose, defined data fields, authentication controls, access limits, audit requirements, error-response standards, credential-expiry and revocation controls, and an incident-management procedure.

Integration status: OneHealth is the only connected system currently in production. Its integration with NovalD is planned for a future release; OneHealth continues to operate on its existing Patient Reference identifiers until that integration is delivered, with the NSN populated per patient as part of the rollout (see the Platform Reference Decisions Log).

Immigration integration: naturalized-citizen registration (Sections 7.5 and 8.3) formally depends on Immigration providing an approval reference. As with OneHealth, this integration is planned and activates when the Immigration system ships; birth-based registration alone is sufficient for NovalD's V1 launch, so V1 does not wait on it.

10. DATA GOVERNANCE, PRIVACY AND SECURITY

NovalD must:

- Encrypt sensitive information in transit and at rest.
- Require multi-factor authentication for privileged users.
- Apply role-based and purpose-based access.
- Mask information that is not required.
- Prevent shared accounts.
- Restrict data exports.
- Separate production, testing and training environments.
- Use fictional or anonymised information outside production.
- Define information-retention and archival periods.
- Conduct privacy and security assessments before onboarding institutions.
- Maintain incident-response procedures.
- Record exceptional access.
- Periodically review user and institutional permissions.
- Protect biometric information with additional controls.
- Prevent ordinary users from deleting identity or audit records.

11. AUDIT AND ACCOUNTABILITY

NovalD must record:

- Successful and failed logins.
- Searches.
- Verification requests.
- Records viewed.
- Applications created.
- Information added or changed.
- Previous and updated values.
- Approvals and rejections.
- Duplicate-review decisions.
- Identity merges and links.
- Evidence accessed.
- Reports and records exported.
- Emergency access.
- User-permission changes.
- Institution and centre changes.
- System-configuration changes.

Each audit event must include: user, role, institution, action, approved purpose, date and time, record affected, outcome, and session or device information where permitted.

12. ANALYTICS AND MONITORING

12.1 National Dashboard

The NICRA National Administrator should see:

- Total registered citizens.
- NSNs issued.
- Birth registrations.
- Naturalized citizen registrations.
- Death registrations.
- Pending applications.
- Rejected applications.
- Late birth registrations.
- Potential duplicates.
- Correction requests.
- Active registration centres.
- Active connected institutions.
- System availability.
- Security and compliance alerts.

12.2 Regional Dashboard

Regional Administrators should see the same operational categories limited to their assigned region.

12.3 Registration Centre Dashboard

- Applications created by the centre.
- Pending reviews.
- Applications returned for more information.
- Approved and rejected applications.
- Average processing time.
- Officer activity.
- Unresolved exceptions.

12.4 Metric Rules

Every metric must define: data source, calculation method, reporting period, update frequency, user roles permitted to view it, and whether personally identifiable information is included.

13. NON-FUNCTIONAL REQUIREMENTS

13.1 Availability

NovalD should provide 99.9% availability for core services. Planned maintenance must be communicated in advance. Critical integrations must have defined fallback responses.

13.2 Performance

- Standard identity-verification requests should return within five seconds.
- Search results should return within an agreed operational threshold.
- The platform must support nationwide concurrent usage.

13.3 Scalability

- The architecture must support growth in citizens, users, centres and connected institutions.
- Services should scale independently where possible.

13.4 Accessibility

- The interface must support users with disabilities.
- Assisted registration must be available.
- Instructions and error messages must use clear language.

13.5 Offline-First and Low-Connectivity Registration

NovalD must support controlled registration in approved rural or low-connectivity centres.

Permitted Offline Activities

Authorised Registration Officers may: begin a new registration, capture demographic information, capture supporting documents, capture photographs and biometrics, save an encrypted registration package, and continue working on a previously saved local draft.

The following actions must not be completed offline:

- Final registration approval.
- NSN generation.
- Duplicate-resolution decisions.
- Material identity corrections.
- Death-registration approval.
- Institutional-access changes.

Local Storage Controls

- Be encrypted on the approved device.
- Be linked to the authenticated officer and registration centre.
- Be stored only on NICRA-approved devices.
- Receive a temporary, globally unique submission reference.
- Be protected from copying to unauthorised storage.

- Be deleted from the local device after confirmed synchronization and the applicable retention period.
- Remain inaccessible after the officer's access is suspended.

Synchronization Process

When connectivity is restored:

1. The device authenticates with NovalD.
2. The queued registrations are uploaded securely.
3. NovalD validates the officer, device and centre.
4. NovalD checks whether each temporary submission reference has already been received.
5. Uploaded records undergo central validation and duplicate checks.
6. Successfully received records receive a central application reference.
7. The device receives confirmation of synchronization.
8. Confirmed local copies are securely removed according to NICRA policy.

Conflict Resolution

Where NovalD identifies conflicting information or another application for the same person: the later submission must not overwrite the earlier record automatically; both submissions are placed in an exception queue; the system shows the conflicting fields; a Verification Officer reviews the records; supporting evidence may be requested; the approved record is retained as the active application; and the rejected, merged or withdrawn submission remains in the audit history.

Failed Synchronization

Where synchronization fails, the encrypted registration package remains in the local queue, the system retries according to configured rules, the officer sees the failure reason and current status, the same submission must not be recreated as a new application, and repeated failures are escalated to technical support, which must not receive unrestricted access to the citizen's readable information.

Offline Audit Requirements

The system must record: the officer who captured the information, the approved device used, the registration centre, the date and time of initial capture, the date and time of synchronization, offline edits made before synchronization, failed synchronization attempts, and conflict-resolution decisions.

13.6 Backup and Recovery

- Identity and audit records must be backed up regularly.
- Recovery procedures must be tested.
- Recovery must preserve record integrity and audit history.

13.7 Technical Architecture

The proposed architecture should separate: identity services, registration services, verification services, access management, audit services, notification services, analytics services, integration services, and biometric services where applicable. As a working assumption for V1, consistent with OneHealth's precedent, the stack is a Next.js frontend with a Firebase/Firestore-style backend; the final technology stack will be confirmed after the functional requirements are approved. Development is expected to use Google Antigravity, an

agentic development platform, as the build environment; this is a tooling choice and does not change the underlying stack.

14. ERROR AND EXCEPTION HANDLING

Scenario	Required Response
Missing mandatory information	Prevent submission and identify the missing fields
Invalid birth notification	Return the notification to the originating facility with the reason
Potential duplicate	Pause NSN issuance and send the application for review
Immigration reference cannot be verified	Pause the application and request confirmation from Immigration
Conflicting parent information	Route the application for enhanced review
Failed connected-system verification	Display a temporary failure and allow controlled retry
Incorrect death registration	Treat as a critical incident and begin urgent reversal
Unauthorised search	Block the request, log it and alert Compliance where required
Failed notification	Retry and place unresolved failures in an operational queue
Interrupted submission	Save a draft where permitted and prevent duplicate submission
Compromised user account	Suspend access and begin security investigation
Citizen disputes an identity decision	Create a formal review or appeal case

Every exception must have an owner, status, priority, action history and escalation route.

15. ASSUMPTIONS AND DEPENDENCIES

- Novaria has a legal basis for national identity and civil registration under two fictional instruments: the Novaria Identity and Civil Registration Act, which establishes NICRA's mandate to register citizens and issue the NSN, and the Novaria Data Protection Act, which governs purpose-based access to identity information and sets penalties for misuse.
- NICRA is authorised to issue and manage the NSN.
- Immigration remains responsible for citizenship decisions.
- OneHealth is available to submit birth notifications from connected health facilities.
- Alternative registration channels exist for births outside connected facilities.
- Connected institutions must complete security and privacy assessments.
- Registration Officers will be trained and authorised.
- Biometric use will be approved before implementation.
- Connected government systems will adopt the NSN as a common identity reference.
- Specialised government records will remain in their responsible systems.

- Novaria's states, zones and LGAs follow the canonical administrative structure defined in Section 20 and the companion Novaria Administrative Reference workbook.

16. RISKS AND MITIGATIONS

Risk	Impact	Mitigation
Unauthorised access	Privacy and trust damage	Purpose-based access, monitoring and sanctions
Duplicate identities	Unreliable registry	Duplicate screening and human review
Incorrect identity changes	Loss of access to services	Evidence, independent approval and appeals
System outage	Disruption across government services	High availability and disaster recovery
Excessive data sharing	Privacy breach	Minimum-information verification
Insider misuse	Identity theft or profiling	Segregation of duties and access monitoring
Exclusion of vulnerable citizens	Incomplete national coverage	Assisted and low-connectivity registration
Poor-quality source information	Incorrect identities	Validation and exception handling
Incorrect duplicate match	Separate persons treated as one	Independent review and citizen appeal
Incorrect death status	Citizen blocked from services	Enhanced validation and urgent reversal
Institutional credential compromise	Large-scale unauthorised access	Rate limits, expiry, monitoring and revocation

17. MINIMUM VIABLE PRODUCT

NovalD V1 will include:

- NICRA user and role management.
- Registration Centre management.
- Institutional onboarding.
- Birth notifications from OneHealth.
- Alternative birth registration.
- Late birth registration.
- Naturalized citizen registration.
- Core citizen identity records.
- NSN generation.
- Identity search and verification.
- Identity corrections.
- Duplicate review.
- Death registration.

- Audit logs.
- Operational dashboards.
- Notifications.
- OneHealth integration.
- Immigration verification integration.
- Core security and compliance monitoring.
- Facial biometric capture and matching.
- Fingerprint capture and matching for eligible citizens.
- Biometric exception management.
- Demographic and biometric duplicate detection.
- Controlled offline registration and synchronization.
- Conflict-resolution queues for offline submissions.
- Citizen self-service (Section 7.16).
- Virtual identity card (view and self-print).
- Basic “Log in with NovalD” authentication for approved government services.

17.1 V1 Exclusions

The following will not form part of V1:

- Police case management.
- Criminal records.
- Missing-person case management.
- Social-service case management.
- Medical records.
- Education records.
- A full digital identity wallet.
- Cross-border identity verification.
- Advanced predictive fraud models.
- Unrestricted citizen editing of identity information.

18. FUTURE SCOPE

Future versions may include:

- Citizen mobile application.
- Digital identity credential.
- Physical identity-card management.
- Mobile registration units.
- Expanded multilingual support.
- Citizen access-history visibility.
- Additional government integrations.
- Advanced fraud analysis with human review.
- Cross-border verification where legally approved.
- Identity services for lawful residents and refugees.

- Online appointment scheduling.
- Expanded citizen self-service.
- Iris biometric capture and matching, subject to future legal, operational and technical approval.

19. REPORTING REQUIREMENTS

Report	Intended User	Purpose
Registration Performance Report	NICRA Administrators	Monitor volumes and processing times
Birth Registration Coverage Report	NICRA and Ministry of Interior	Monitor registration coverage
Late Birth Registration Report	NICRA Operations	Monitor delayed registrations and outcomes
Duplicate Review Report	Duplicate and Compliance Teams	Track possible duplicates and decisions
Identity Correction Report	NICRA Operations	Monitor correction volumes and outcomes
Institutional Access Report	Compliance	Review searches and verification activity
Security Exception Report	Security and Compliance	Investigate suspicious activity
User Access Review Report	NICRA and Institution Administrators	Review active users and permissions
Platform Availability Report	Technical Operations	Monitor uptime and failures
Death Registration Report	NICRA Operations	Monitor death notifications and approvals

20. ADMINISTRATIVE STRUCTURE OF NOVARIA

This section defines Novaria's states, zones and LGAs. It is the canonical version for NovalD, OneHealth and any future Novaria product; see the companion workbook **Novaria Administrative Reference.xlsx** for the full, extensible dataset.

Novaria is organised into five geopolitical zones, nine states, and one federal capital territory. Each state is divided into Local Government Areas (LGAs); the Novaria Capital Territory (NCT) is divided into five Area Councils. NovalD's registration centres, dashboards and address fields (Section 7.6 and 7.17) will use this structure.

20.1 Zones and States

Zone	States	Capital(s)
North	Kandova State, Arinana State	Kandova, Arinana
South	Manoca State, Sorima State	Manoca, Sorima
East	Tronto State, Mivo State	Tronto, Mivo
West	Corintha State, Ostravo State	Corintha, Ostravo
Central	Kolvira State	Kolvira
Federal (no zone)	Novaria Capital Territory (NCT)	Novaria City

20.2 States and Local Government Areas

State	Code	LGAs
Kandova State	KD	Kandova, Varanta, Tavren, Birova, Lekori, Mirova, Jorato
Arinana State	AR	Arinana, Vol, Shalo, Madilla, Dalmin
Manoca State	MN	Manoca, Franc, Kessari, Minka, Avana
Sorima State	SR	Sorima, Solovivi, Ishono, Tashara, Vornis
Tronto State	TR	Tronto, Ormari, Kelmora, Denkora, Halvot
Mivo State	MV	Mivo, Petrona, Bratona, Rokova, Ternis
Corintha State	CR	Corintha, Selkiv, Dorvana, Milante, Vashira
Ostravo State	OS	Ostravo, Stenira, Boverin, Yelmira, Fintara
Kolvira State	KV	Kolvira, Astrivo, Kelvano, Bralinda, Torvesk

20.3 Novaria Capital Territory (NCT)

The NCT is the federal capital district and the seat of NICRA headquarters and the Ministry of Interior. It sits outside the five zones; in the same way Section 5 keeps NICRA's national authority separate from regional administration. It is divided into five Area Councils: North, South, East, West and Central Area Council.

20.4 How this was built, and why it matters for OneHealth

Kandova State's seven LGAs, and several of the other LGA names, are already in use in OneHealth's live sample data; this reference formalizes them rather than replacing them. Two things are new: the label "North Region" (used loosely in some OneHealth exports) is now formally Kandova State, and the directional LGA values already seen under "Capital Territory" rows (North, South, East, West, Central) are now formally the NCT's Area Councils.

This means no correction is required in OneHealth's existing patient data before NovalD can reference it; only the display label for one state name is affected. A short mapping table for that future cleanup pass is included as a separate sheet in the reference workbook, so it does not need to be repeated here.

20.5 Extending the structure

Additional states, LGAs, or zone reassignments should be added to the Novaria Administrative Reference workbook directly, using the same code pattern (next unused 2-letter state code, LGA codes as STATECODE-01, -02, ...), so every product that reads from it stays in sync automatically.

21. OPEN DECISIONS

NovalD as the product name, biometric scope for the MVP, and the general structure of the NSN are already decided (Sections 1, 7.8 and 7.9). The administrative structure is resolved in Section 20. The following items are now resolved:

- 1.** Check-digit algorithm: the NSN's twelfth digit uses the Luhn (mod-10) algorithm.
- 2.** Biometric vendor and matching technology: no specific commercial vendor is named in this document; biometric templates and matching must conform to ISO/IEC 19794 (fingerprint) and ISO/IEC 39794 (facial) interchange standards, with vendor selection made during procurement.
- 3.** Biometric matching thresholds: the three-tier confidence model (low, medium, high; see Section 7.14) is decided. The specific numeric thresholds remain open and will be set during biometric testing, since they depend on the vendor and data quality.
- 4.** Biometric update intervals: children's biometrics are recaptured at ages 12 and 18 in addition to the initial capture at age five (see Section 7.8). Adults recapture biometrics at intervals set by NICRA, nominally every ten years, or sooner if requested.
- 5.** Physical identity cards: not included in V1. V1 provides a virtual identity card through citizen self-service (see Sections 7.16 and 17); physical cards are future scope.
- 6.** Citizen self-service: included in V1 (see Sections 7.16 and 17).
- 7.** Statutory registration period: a birth must be registered within 60 days of the date of birth; a registration submitted after that period is treated as a late registration (see Section 7.4).
- 8.** Evidence for births outside a health facility: a signed statement from the attending health worker or traditional birth attendant, or, where none was present, a sworn affidavit, together with witness or community verification where available (see Section 7.3).
- 9.** Evidence for late registration: at least one corroborating document, such as a school enrolment record, a religious or baptismal record, or a sworn affidavit of age declaration, in addition to the standard registration evidence (see Section 7.4).
- 10.** Evidence for material corrections: evidence appropriate to the specific change, for example a court order, marriage certificate or government gazette notice for a legal name change, or medical or school records plus a sworn affidavit for a date-of-birth correction, reviewed case by case by a Verification Officer (see Section 7.13).
- 11.** Identifier for lawful residents and refugees: a distinct identifier, the Novaria Resident Number (NRN), not the NSN (see the new Section 7.20).
- 12.** Governing legislation: the fictional Novaria Identity and Civil Registration Act (establishing NICRA's mandate and the NSN) and the Novaria Data Protection Act (governing access and misuse penalties); see Section 15.
- 13.** Technology stack: a Next.js frontend with a Firebase/Firestore-style backend is the working assumption for V1, consistent with OneHealth's precedent, and will be refined during technical design (see Section 13.7). Development tooling: Google Antigravity is the expected build environment; it does not change the stack itself.
- 14.** Implementation timeline: a phased rollout, with Phase 1 covering core registration, NSN issuance and OneHealth integration, and Phase 2 covering biometric expansion and additional agency integrations; specific dates will be set during project planning.

22. ACCEPTANCE CRITERIA

This section starts the Acceptance Criteria for NovalD, covering the highest-risk MVP flows first: NSN issuance, birth registration, duplicate handling, death registration, and Log in with NovalD. It should grow alongside Section 7 rather than being written all at once at the end.

22.1 NSN Generation and Uniqueness (Section 7.9)

ID	Given	When	Then
AC-1.1	A registration has received final approval and has no existing NSN	The system issues an NSN	A unique 12-digit NSN with a valid check digit is generated, and it is not manually editable
AC-1.2	A generated NSN would collide with an existing NSN	The collision is detected during issuance	The number is rejected and regenerated before being assigned
AC-1.3	A citizen has forgotten their NSN and completes identity verification successfully	They request their NSN	The existing NSN is retrieved; no new NSN is created

22.2 Birth Registration from OneHealth (Sections 7.1, 7.2, 8.1)

ID	Given	When	Then
AC-2.1	A connected hospital submits a complete birth notification	NovalD receives it	A unique birth-notification reference is generated and returned to OneHealth; no NSN is issued yet
AC-2.2	A birth notification is missing a required field	NovalD validates it	The notification is placed in an exception queue and not silently accepted
AC-2.3	A Registration Officer submits a civil registration application built from a valid notification	A Verification Officer and, where required, a Senior Approver approve it	The identity record is created and an NSN is generated only after final approval

22.3 Duplicate Detection (Sections 7.14, 8.5)

ID	Given	When	Then
AC-3.1	A new registration matches an existing record above the medium-confidence threshold	The match is detected	NSN issuance is paused and the case is routed to a Duplicate Review Officer, not auto-resolved
AC-3.2	A Duplicate Review Officer recommends a merge	A Senior Approver independently reviews the recommendation	Both original records and the full decision history are preserved regardless of

ID	Given	When	Then
			outcome
AC-3.3	A citizen believes they were incorrectly matched as a duplicate	They dispute the match	A formal review case is created and the citizen's application is not rejected without it

22.4 Death Registration and Reversal (Sections 7.15, 8.6)

ID	Given	When	Then
AC-4.1	An approved source submits a death notification with supporting evidence	An authorised officer approves it	The citizen's status changes to deceased; the NSN and identity record are preserved, not deleted
AC-4.2	A citizen has been incorrectly marked as deceased	The error is reported	The record supports urgent reversal and the incident is treated as critical, not as a routine correction

22.5 Log in with NovalD (Sections 7.10, 8.7)

ID	Given	When	Then
AC-5.1	A citizen selects "Log in with NovalD" on an approved government platform	They authenticate successfully	Only the identity attributes the platform is approved to receive are returned; never the citizen's password, PIN or biometric data
AC-5.2	A platform requesting authentication is not on NICRA's approved list	A login is attempted	The request is rejected before any identity attribute is returned

23. APPENDICES

The following appendices will provide detailed implementation references after the main requirements are approved. Acceptance Criteria (started in Section 22) and the Novaria administrative structure (Section 20, with the full dataset in the companion workbook) are no longer on this list, since both now exist.

- Data Dictionary.
- NSN Format and Validation Rules.
- Full Role and Permission Matrix.
- Registration Status Definitions.
- Identity Update Classification.
- Duplicate Resolution Framework.
- Integration Field Matrix.
- Notification Catalogue.
- Audit Event Catalogue.
- Report Definitions.
- Test Scenarios.
- User Flows.
- System Architecture Diagram.

Working Draft Note: This version remains open for review and editing. It will only be converted into a final, numbered release after all remaining open decisions (Section 21) have been agreed.